

Discrete Equalization Audits from Logs: Menu-Level Finite-Sample TV Certificates for Receipt-Grade Claims

February 28, 2026 (archive v150)

Abstract

Many papers in this archive reduce anonymity claims to an *equalization contract*: for a declared interface projection X , the induced distributions P_k are close across keys/classes/policies k . When only logs are available, deployments need a referee-grade way to *certify* such closeness without rederiving statistics in every mechanism paper.

This note centralizes a conservative, discrete finite-sample audit recipe: a menu-level bound (with many-testing correction) that upgrades empirical frequencies to a simultaneous total-variation guarantee, and records the needed parameters in a receipt-facing evidence object.

1 Setup: discrete projections and “menus” of claims

Fix a discrete projection $X \in \{1, \dots, m\}$ (bucketed timings (see Synthesis 27 for bucket design), [7] retry-pattern hashes, prefix-length bins, ...) that an exposure normal form (ENF) commits to. [1]

A deployment typically audits not one claim but a *menu*: many keys/policies/classes $k \in \mathcal{K}$ (and sometimes multiple projections), often with different sample sizes. This note treats the menu as a finite index set $\mathcal{M}$. For each menu item $\mu \in \mathcal{M}$ and each class $k \in \mathcal{K}(\mu)$, collect $n_{k,\mu}$ i.i.d. samples of X and form an empirical distribution $\hat{P}_{k,\mu}$.

Definition 1 (Menu-level statistical confidence). *A menu-level statistical failure probability is a single parameter $\delta_{\text{stat}} \in (0, 1)$ intended to hold simultaneously for all (k, μ) in the declared audit scope.*

2 A conservative TV radius bound

The following standard bound (Weissman et al.) controls the L_1 deviation of multinomial empirical distributions. [9]

Lemma 1 (Weissman L_1 deviation $\Rightarrow$ TV radius). *Let P be a distribution on $\{1, \dots, m\}$ and let $\hat{P}$ be the empirical distribution of n i.i.d. samples from P . Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\|\hat{P} - P\|_1 \leq r(n, m, \delta) := \sqrt{\frac{2}{n} \log\left(\frac{2^m - 2}{\delta}\right)}.$$

Equivalently, $\text{TV}(\hat{P}, P) \leq r(n, m, \delta)/2$.

Remark 1 (Why this is “conservative”). *The bound is uniform and simple to publish. Sharper alternatives exist (e.g., profile maximum likelihood, Bernstein-type refinements), but this note prioritizes a short, receipt-grade certificate that does not depend on opaque tooling.*

3 Menu-level equalization certificates

A union bound upgrades Lemma 1 to a simultaneous menu-level certificate.

Corollary 1 (Simultaneous TV radii over a menu). *Fix $\delta_{\text{stat}} \in (0, 1)$. For each audited pair (k, μ) define*

$$\delta_{k,\mu} := \delta_{\text{stat}}/|\mathcal{I}|, \quad r_{k,\mu} := r(n_{k,\mu}, m_\mu, \delta_{k,\mu}),$$

where $|\mathcal{I}|$ is the number of audited pairs (k, μ) and m_μ is the alphabet size of the projection for menu item μ . Then with probability at least $1 - \delta_{\text{stat}}$ we have $\text{TV}(\hat{P}_{k,\mu}, P_{k,\mu}) \leq r_{k,\mu}/2$ simultaneously for all audited (k, μ) .

From radii to equalization. For two audited classes (k, μ) and (k', μ) on the same projection,

$$\text{TV}(P_{k,\mu}, P_{k',\mu}) \leq \text{TV}(\hat{P}_{k,\mu}, \hat{P}_{k',\mu}) + \frac{r_{k,\mu} + r_{k',\mu}}{2}$$

holds on the same $1 - \delta_{\text{stat}}$ event. This is the simplest log-only route to an equalization contract that can be converted to odds inflation (Anonymity A) and profiling budgets (Anonymity B). [4, 5]

4 Receipt-facing publication (minimal fields)

A deployment should not publish the full logs. Instead, publish a compact `profiling_evidence` object that commits to:

- the projection identifier (ENF-ID or the discrete projection name/hash),
- the menu scope (which keys/policies/classes were covered),
- the statistical parameter δ_{stat} and the per-item allocation rule (e.g. Bonferroni),
- sample sizes $n_{k,\mu}$ and observed empirical distances $\text{TV}(\hat{P}_{k,\mu}, \hat{P}_{k',\mu})$ (or a derived bound),
- the resulting certified TV bound(s) used by the receipt line items. [2]

For a concrete instantiation with digest-bound bundles and replay hooks, see the worked example receipts in Synthesis 17. [3]

5 Research warranted (short list)

- **Better finite-sample bounds with the same surface:** replace the uniform Weissman radius with a tighter plug-in bound while keeping a single published δ_{stat} .
- **Adaptive menus (change control over time):** centralize the “menu grows, audits repeat” bookkeeping via an anytime spending schedule (Synthesis 26) rather than re-deriving it per paper.[6]
- **Large alphabets:** when a projection induces a large alphabet size m , coarsening/partition design should be treated as a declared, receipt-facing map rather than an implicit reviewer assumption; see Synthesis 28.[8]

References

- [1] Anonymous. Trace Interfaces for Anonymous DHT Lookups: Exposure Normal Forms and Bounded Projections. Synthesis 3, The-One-True-Archive (2026).
- [2] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
- [3] Anonymous. Worked Example: Receipt/Artifact Interlock for OHTTP + DHT Profiling (with optional variants). Synthesis 17, The-One-True-Archive (2026).
- [4] Anonymous. Odds Inflation as a Universal Anonymity Contract. Anonymity A, The-One-True-Archive (2026).
- [5] Anonymous. Anonymous DHT Profiling and the Equalization Contract. Anonymity B, The-One-True-Archive (2026).
- [6] Anonymous. Anytime Statistical Spending for Menu Audits: Change-Control-Safe Confidence Budgets. Synthesis 26, The-One-True-Archive (2026).
- [7] Anonymous. Bucket Design for Continuous Timing: Safe Discretization Rules. Synthesis 27, The-One-True-Archive (2026).
- [8] Anonymous. Coarsening and Large-Alphabet Projections for Log Audits: Safe Partitioning Rules that Keep Certificates Sample-Feasible. Synthesis 28, The-One-True-Archive (2026).
- [9] T. Weissman, E. Ordentlich, G. Seroussi, S. Verdú, and M. J. Weinberger. Inequalities for the L_1 deviation of the empirical distribution. *Hewlett-Packard Labs Technical Report*, 2003.